A width-sensitive affine moment bound for Dirichlet L-functions

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16 October 2026

Abstract

We sharpen the all-moduli Dirichlet nonvanishing boundary from 0.874840 to 0.874836. The new ingredient is a joint estimate that preserves the total logarithmic width of the selected primes through both allocation stages and couples the resulting saving to a valuation constraint on the actual numerator. Keeping this moving width, rather than its smallest admissible value, gives a stronger affine moment region. Combining the estimate with the existing bounds for the remaining source and exceptional terms yields the moment inequality needed for the zero-free region.

1 Result and proof strategy

Theorem 1. *For every positive integer n, every Dirichlet character χ modulo n, and every $z \in \mathbb{C}$,*

$$\Re z > \vartheta := \frac{218709}{250000} = 0.874836, \qquad \neg(\chi = 1 \text{ and } z = 1) \quad \Longrightarrow \quad L(\chi, z) \neq 0. \tag{1}$$

The analytic framework is inherited from OpenAI and Collet [1, 2]. It studies a global zero supremum β in an Eisenstein Hecke family and transfers its bound to all Dirichlet characters. We use the established $\beta \le 0.874840$ [6] as an input. The new estimate improves the moment calculation inside that argument; it does not replace the underlying zero-density or continuation theorems.

Here is the moment region that the argument needs. Normalize all logarithmic lengths by $\log U$, where $U > 1$ is the source scale. The two ordinary Dirichlet-polynomial lengths are m_1, m_2, and $s = \sum_j \log \mathrm{N}(P_j)/\log U$ is the total width of the selected prime ideals P_j. Write $\mathcal{E}_\eta(\Phi, \mathbf{p})$ for the normalized energy in Collet's initial affine moment, with fixed character η, shifts of size at most T, and the original masks and selected-prime weights.

Proposition 1 (Affine moment region)**.** *For every $\varepsilon > 0$, the initial energy satisfies*

$$\mathcal{E}_\eta(\Phi, \mathbf{p}) \ll_{\eta,\varepsilon} \mathcal{D}(\Phi)\mathcal{S}(\mathbf{p})^2(1+T)^N U^{1+\varepsilon} \quad \text{for} \quad m_1 + m_2 + \lambda s \le B, \tag{2}$$

where

$$\lambda = \frac{19793}{5000}, \qquad B = \frac{987047}{1000000}, \qquad \frac{41}{100} \le m_i \le \frac{11}{25}, \quad \frac{17}{500} \le s \le \frac{1}{25}.$$

Here Φ is a nonnegative radial weight, $\mathbf{p}$ a pair of test functions, $\mathcal{D}$ the standard diagonal control, $\mathcal{S}$ a finite Schwartz control, and N a finite degree. The prime-factor partition is sufficiently fine, as in the initial-moment theorem.

The significance of (2) is its available prime width: for equal plain lengths m, it permits $s \le (B - 2m)/\lambda$. This width later enters the count of detector rows. A saving lost at the moment stage therefore cannot be recovered merely by changing the detector parameters.

The source decomposition has a LOW part handled by the incumbent moment, and a HIGH part requiring marked counting. The new part is a width-sensitive estimate for the latter. After the two source allocations, some selected width has entered common factors, some is frozen, and the rest survives. We keep their sum tied to the original width and use the numerator valuation bound before separating these contributions. Section 2 proves the resulting joint inequality. Section 3 combines it with the known LOW estimate and the error terms to prove Proposition 1. Section 4 applies this moment to the detector and weighted numerator, then invokes the inherited continuation argument to prove Theorem 1.

2 The joint HIGH estimate

The numerator and the common factors

Let ν denote the auxiliary density level of a HIGH source bin; its zero-real-part threshold is $(1+\nu)/2$. Put

$$\rho(\nu) = 1 - \frac{10\nu}{3(5-2\nu)}, \qquad \gamma(\nu) = \nu - \max\left(4\rho(\nu) - 1, \frac{2+22\rho(\nu)}{21}\right). \tag{3}$$

Here ρ is the exponent supplied by the unmarked source estimate. In the range used below, $1/4 \le \rho \le 2/5$.

The first allocation has logarithmic ideal lengths c, d, active radical width R, cancelled radical width R_0, and numerator width L_c. The nonnegative quantity E is its inactive-projector loss. These quantities belong to the same allocation. Besides $R \le \min(c,d)$ and $R_0 \ge 0$, the actual numerator satisfies

$$L_c \le 5(R - R_0), \qquad 4L_c \le 5(c+d). \tag{4}$$

The second inequality follows prime by prime from the numerator valuations and holds in either reflection orientation. Its use together with the first inequality, rather than as an independent bound on a free numerator, is the relevant coupling.

For $x_+ = \max(x, 0)$, the common-factor part of the HIGH exponent is

$$\mathcal{F} = (1-\rho)(c-d)_+ - \nu c - (1+\rho)R - E - \frac{(3c-5d-R)_+}{6} + \rho L_c + R_0.$$

We obtain

$$\mathcal{F} \le -\gamma(\nu)c - E. \tag{5}$$

For completeness, when $c \le d$, insert $L_c \le 5(R - R_0)$ and $R \le c$; this gives rate $\nu - (4\rho - 1)$. When $d \le c$, a positive linear combination of (4) gives

$$21\rho L_c + 21R_0 \le (19\rho - 4)(c+d) + (16 + 29\rho)R.$$

Insert this and retain the positive-part penalty in $\mathcal{F}$; the resulting rate is $\nu - (2+22\rho)/21$. Taking the smaller of these two rates proves (5). The two-piece rate is essential to cover both orderings.

Width survives both allocations

Let $u, f, v \ge 0$ be the common width removed by the first allocation, the frozen width in the second, and the surviving selected width. Let Q be the remaining logarithmic frequency scale. The source identities give bounds of the form

$$u + f + v \ge S, \qquad Q \ge F - 2u. \tag{6}$$

The common width pays rate $\gamma = \gamma(\nu)$ by (5), the second allocation pays rate ν, and the marked inverse/plain estimate of timbalud [4] pays

$$r(\nu)\min(v, \tau_s Q), \qquad r(\nu) = \frac{32}{39}\frac{\nu(5-6\nu)}{5-2\nu}, \quad \tau_s = \frac{13}{400}.$$

Thus the joint payment is $\gamma u + \nu f + r\min(v, \tau_s Q)$.

Lemma 1 (Conserved-width payment)**.** *If $S, F \ge 0$, $0 \le \gamma \le \nu$, $r \ge 0$, $\tau_s \ge 0$ and $2r\tau_s \le \gamma$, then (6) implies*

$$\gamma u + \nu f + r\min(v, \tau_s Q) \ge \min\{\gamma S,\ rS,\ r\tau_s F\}. \tag{7}$$

If $v \le \tau_s Q$, compare the three width rates and use $u + f + v \ge S$. Otherwise use $Q \ge F - 2u$: the coefficient of u is $\gamma - 2r\tau_s \ge 0$, leaving at least $r\tau_s F$. The large sixth-power tail has payment $\gamma u + \nu f + tQ$, where $t = (6\rho - 1)/250$; the same proof gives tF when $2t \le \gamma$. These inequalities keep the common and surviving widths in one estimate. Replacing S by its minimum prematurely discards saving precisely where the amplifier and total source volume are also increasing.

3 Completing the affine moment

We now choose the amplifier while keeping the width variable. Let b be the upper endpoint of the cell containing s, with $b - 10^{-5} \le s \le b$; thus $3401/100000 \le b \le 1/25$. Write $a(b)$ for the amplifier exponent and $A(b)$ for the total source-volume exponent. Take

$$q = \frac{839}{1250}, \qquad \kappa_0 = 2(0.874840) - 1 = \frac{9371}{12500}, \qquad r_L = \frac{1}{4000000},$$
$$D(b) = B - 1 + (6\kappa_0 - \lambda)b, \qquad a(b) = \frac{qD(b)}{6(\kappa_0 - q)} + \frac{r_L}{1 - q/\kappa_0},$$
$$A(b) = B - (\lambda - 1)(b - 10^{-5}), \qquad S(b) = b - 10^{-5} - 10^{-3}, \qquad F(b) = \frac{122}{125}\big(1 - 6a(b) - 10^{-3}\big). \tag{8}$$

The formula for A follows from $m_1 + m_2 + s \le B - (\lambda - 1)s$. The two allocations and the discarded tiny slots give (6) with $S = S(b)$ and $F = F(b)$. In particular the supply varies with the original width s.

LOW. The established incumbent is κ_0, not the new value $2\vartheta - 1$ that we seek to prove. The exact identity

$$a(b) - \frac{q}{6\kappa_0}\big(D(b) + 6a(b)\big) = r_L \tag{9}$$

pays the complete LOW removal cost. Both its capacity and removal price use the same incumbent. The original positive/zero moment estimates and height bounds therefore apply without circular feedback. The LOW/HIGH zero-bin junction is $(1 + q)/2 = 2089/2500$.

HIGH. Before the width payment, the remaining saving is

$$H(b, \nu) = 1 - \rho(\nu) - \nu A(b) - a(b)(5 - 6\rho(\nu)).$$

Combining this with (7) and the tail estimate reduces HIGH to four comparisons:

$$H + \gamma S - \tfrac{1}{20000} - \tfrac{1}{200000} \ge 10^{-6}, \qquad H + rS - \tfrac{1}{20000} - \tfrac{1}{200000} \ge 10^{-6},$$
$$H + r\tau_s F - \tfrac{1}{20000} - \tfrac{1}{200000} \ge 10^{-6}, \qquad H + tF - \tfrac{1}{20000} - \tfrac{1}{200000} \ge 10^{-6}. \tag{10}$$

The $1/200000$ term reserves the raw-root charge in the marked estimate; the unmarked common-width estimate can omit it. These inequalities hold for every b in its cell range and $q \le \nu \le 7/10$. At $\nu = q$ each is affine in b, so rational arithmetic at the two endpoints suffices. To extend in ν, H increases by at least $(\nu - q)/5$, γ is nondecreasing, and r, t decrease by at most $\nu - q$ and $(\nu - q)/20$, respectively. Use $S, \tau_s F \le 1/25$ and $F \le 1$. For $7/10 \le \nu \le 3/4$ the unmarked estimate already gives $H \ge 10^{-6}$.

ERROR and the remaining terms. Error deletion retains the original $3/4$ incumbent and hence the coefficient $9/2$, independently of (9). Its defect and first-branch saving are

$$D_E(b) = B - 1 + (9/2 - \lambda)b, \qquad a(b)/4 - D_E(b)/6 \ge 1/1000. \tag{11}$$

The sixth and seventh deletion branches have separate positive reserves. All three exceed the pre-loss reserve $7/30000000$; after branch-loss absorption the normalized physical sum uses saving $7/60000000$, with allocation and shell losses still separate. The moving-prime average is paid once and the retained radical witness is preserved. Finally $1 - A(b) - 5a(b) \ge 227876729/3446640000$ pays the zero term; the amplifier coefficient is at most 2048, and the inherited tail estimates complete the sum. The original source decomposition, with both reflections and all shifted heights, now proves Proposition 1.

4 From the moment to the nonvanishing boundary

Let δ now denote the *outer detector* density; it is independent of the inner source parameter ν in Sections 2–3. For inverse length r, plain length m, cutoff $t = r + m$ and normalized amplitude $p \in [0, 1/2]$, the three count exponents are

$$I = 1 - \delta\big(p + (1-p)r\big), \quad P = 1 - 2\delta m - 2\delta p\frac{B - 2m}{\lambda}, \quad L = 1 - \delta + (5/6 - \delta)(t - 1).$$

The new width in P is exactly Proposition 1. Balancing I, P, L gives

$$D_p = 3 - p - 4p/\lambda, \quad a_p = 2 - 4p/\lambda, \quad c_p = 1 - 2pB/\lambda,$$
$$J = (5/6 - \delta)D_p + \delta(1-p)a_p, \qquad R_{\text{aff}} = 1 - \delta + \frac{(5/6 - \delta)\delta(1-p)c_p}{J}. \tag{12}$$

Following DaysSky [5], apply the moment separately to each character in the finite detector dictionary; maxima of the degrees and unions of the seminorms give a common height range for that dictionary. Use (12) on $[.451, .49] \times [.49, .5]$, the accepted affine count on $[.445, .451] \times [.4975, .5]$, and the original count elsewhere. The selected widths and plain lengths remain within the corresponding moment windows; in particular $m \le 847/2000$ on the new affine branch.

Choose the outer geometry

$$h = 3181/4000, \quad d_- = h - 3/5000, \quad d_+ = h + 10^{-10}, \quad \ell = 11/3 - 4\vartheta, \quad x = 1 + \ell - h, \quad y = h - 2\ell, \quad \kappa_* = 2\vartheta - 1.$$

The lower and upper height ranges use capacities

$$C_O(d) = \frac{\min(d, 2y - d)}{6\kappa_*}, \qquad C_N(d) = \frac{2y - (2 - B)d}{\lambda}.$$

Writing R for the chosen count exponent, the weighted-numerator comparison is

$$E(d, C, \delta, p) = \frac{1+\delta}{2} + \frac{13h}{75} - \frac{y}{2} - \frac{\ell}{2} + \ell\delta p + d\left(\frac{1}{4} - \frac{17}{50}\right) - \frac{\delta pC}{2} + \frac{3dR}{4} \;\le\; \vartheta - 10^{-7}. \tag{13}$$

This holds on the full rectangle $0 \le \delta \le 3/4$, $0 \le p \le 1/2$. After multiplying by the positive count denominators, rational Bernstein polynomials prove (13). Affinity in d leaves five ends: $1/200, y, d_-$ with C_O, and d_-, d_+ with C_N. The shifted source uses $t_0 = 103/100000$; its large-shift and tail estimates are included. Feedback inequalities extend the comparison from κ_* to the actual $\kappa = 2\beta - 1$.

Levy's weighted numerator and Collet's continuation argument [3, 2] now compare the lower and upper bounds for the same normalized probe. If $\beta > \vartheta$, choose a common positive continuation reserve and $\omega = (\beta - \vartheta)/2$ before the target characters. The finite-dictionary moments above allow target-dependent height ranges, while (13) supplies a common positive gap. Continuation then strictly improves the global supremum β, a contradiction. The inherited Hecke-to-Dirichlet transfer proves (1).

Acknowledgments. We build on OpenAI's framework, Collet's source and continuation theory, Levy's weighted numerator, timbalud's asymmetric marked counts, and DaysSky's affine detector. Related affine-count arithmetic by liqsweep/blueberryvertigo is acknowledged. Zhongpai Gao used AI assistance. Predecessor source notices and licenses are retained; this manuscript is released under Apache-2.0 subject to those notices.

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